MILITARY SPECIFICATION

PROJECTILE, 105MM, TP-T, M489
METAL PARTS ASSEMBLY

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for the fabrication of parts, assembly and packing of one type of metal parts for TP-T, M489 Projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-60536 - Coating, Anodic, Hard, For Aluminum and Aluminum Alloys

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

PRODUCT AND PACKAGING DRAWINGS

7548604 - Pallet (Boxed Type) for Projectiles and Projectile Metal Parts
10523473 - Projectile, 105MM, TP-T, M489 Metal Parts Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

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ASTM Methods - American Society for Testing and Materials

E8	-	Tension Testing of Metallic Materials
B117	-	Salt Spray (Fog) Testing
A322	-	Steel Bars, Alloy, Standard Grades
D785	-	Test for Rockwell Hardness of Plastics and Electrical Insulation Materials
D792	-	Test for Specific Gravity and Density of Plastics by Displacement
D2290	-	Test for Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawing, referenced specifications and the following:

3.2 Heat treatment or stress relief. The heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable component drawing.

3.2.1 Hot forge - heat treatment (if applicable). Bodies manufactured by this method shall be subjected to heat treatment to attain required mechanical properties. The heat treatment shall consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit (F), followed by an oil quench. After quenching, the body shall be tempered without delay at a temperature above 750 degrees F., suitable to produce required mechanical properties. After removal from tempering furnace, the projectile body shall be slowly cooled in air (forced air permitted) to a temperature of not more than 700 degrees F., after which cooling may be accelerated by water spray or similar means.

3.2.2 Cold shaping (if applicable). Cold shaping shall be so accomplished as to produce in the body the mechanical properties specified. The body shall be stress relieved at a temperature of 700 degrees F., minimum, following the final forming operation

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represented by the tensile samples. At the option of the

during which time plastic flow takes place. The minimum soak time at or above minimum temperature shall be such as to achieve stress relief and still meet mechanical properties specified on the applicable drawing.

3.3 Protective coating. Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration. Salt spray testing shall be in accordance with the applicable drawings.

3.4 Painting. Painting shall comply with the requirements on the applicable drawing. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.5 Functional

3.5.1 Metal parts security. There shall be no loss, breakup, loosening or deformation of any metal parts component, except the obturating band and seal, in the gunbore or in flight, during any of the proving ground tests conducted.

3.5.2 Target-impact-dispersion. The pooled probable error shall not be greater than .22 mils vertical and .24 mils horizontal when tested in accordance with paragraph 4.5.4.2. See 6.5 for method of calculating pooled P.E.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.7 Torque requirements.

3.7.1 Assembly torque. Threaded joints shall withstand subsequent reapplication of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.7.2 Disassembly torque. Threaded joints shall withstand application of the minimum disassembly torque specified on the applicable drawing, without moving, after thread sealing compound (if required) has dried.

3.8 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

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3.8.1 Material defects. All components shall be free of porosity and excessive flash.

3.8.2 Metal defects. All components, and component cavities shall be free from cracks, splits, cold shuts, inclusions, pipes, porosity, bursts, folds, seams, and other metal defects.

3.8.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.8.4 Burr. No component shall have a burr.

3.8.5 Foreign matter. All parts and assemblies shall be free of dirt, chips, oil, grease, rust, corrosion or other foreign matter.

3.8.6 Molding defects. Soundness of the material shall be in accordance with the applicable drawing and specification. All molded components shall be free of blisters, cracks, check, inclusions, scratches, voids, and any surface or internal defect which may detrimentally affect serviceability or appearance in the judgement of the appropriate technical agency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Nose Cap, or	10523477	20
Collar Ring	9335527	20
Spike, or	10523476	20
Spike for Swagged Nose Collar	9331315	20
Spike Assembly	9331316	20
Fin	8597610	20
Base	10523474	20
Fin and Base Assembly	9327365	20
Obturator	9327293	100
Seal	9327359	100
Body	10523475	20
Body and Seal Assembly	9327295	20
Projectile Metal Parts (Prior to Paint)	10523473	20
Projectile Metal Parts (After Paint)	10523473	55
Test Panels - See 4.3.4		

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile Metal Parts, M489				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
I.	Components and Sub-Assemblies				
	1. 10523477 - Cap, Nose or	20		3.1	4.4.2.1
	9335527 - Collar Ring	20		3.1	4.4.2.2
	2. 10523476 - Spike or	20		3.1	4.4.2.3
	9331315 - Spike (For Swagged Nose Collar)	20		3.1	4.4.2.4
	3. 9331316 - Spike Assy.	20		3.1	4.4.2.5 or 4.4.2.6
	4. 8597610 - Fin	20		3.1	4.4.2.7
	5. 10523474 - Base	20		3.1	4.4.2.8
	6. 9327365 - Fin and Base Assy.	20		3.1	4.4.2.9
	7. 9327293 - Obturator	100		3.1	4.4.2.10 (See Note 1)
	8. 9327359 - Seal	100		3.1	4.4.2.11 (See Note 2)
	9. 10523475 - Body	20		3.1	4.4.2.12
	10. 9327295 - Body and Seal Assy.	20		3.1	4.4.2.13
II.	Projectile Metal Parts (Prior to Paint) - 10523473 Examination for Defects and Tests	20		3.1	4.4.2.14
NOTE 1.	After the dimensional and visual inspection of the 100 samples is completed, subject 60 samples (20 for each temp.) to the ultimate load and extension test. The remaining 40 samples shall be subjected to the hardness test. After the hardness test is completed, subject 20 samples to the moisture content and water absorption test.				
NOTE 2.	Same as Note 1 with the exception that the hardness will not be conducted.				
NOTES:					

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TABLE I - First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile Metal Parts, M489				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
III.	Projectile Metal Parts (After Paint) - 10523473				
	1. Examination for Defects and Tests	55		3.1	4.4.2.15 (See Note 3)
	2. Ballistic tests				
	a. Target-Impact-Dispersion and Security	25	4.4.3.3.3 & 4.4.3.3.4	3.5.1 & 3.5.2	4.5.4.2
	b. Excess pressure - security	25	4.4.3.3.3	3.5.1	4.5.4.1

NOTES: 3. After the dimensional and visual inspection of the 55 samples is completed, subject three samples to the salt spray test and two to the adhesion of paint test. The remaining 50 shall be subjected to the ballistic tests.

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4.3.4 Abrasion resistance.

4.3.4.1 Test panels. When the protective coating is being applied to the first article sample of the fin and base assembly (or components thereof), 4 test panels as specified in paragraph 9.4.3 of MIL-C-60536 shall be processed along with the product, through each step in the coating process. After coating, the panels will be forwarded to a government approved laboratory for testing in accordance with 9.4.3 of MIL-C-60536 to determine compliance with paragraph 8.4.1 of MIL-C-60536.

4.3.4.2 File test. At the same time the test panels provided by 4.3.4.1 are being subjected to the abrasion test at the Government approved laboratory, 2 fin and base assemblies (or two sets of component parts thereof) from the first article sample shall be subjected to the file test in accordance with paragraph 9.4.4 of MIL-C-60536 to establish correlation (of hardness of coat) between the panels from 4.3.4.1 and the file test of the fin and base assembly or components thereof. The file test will be performed using the equipment and procedures that will be used in production testing.

4.3.4.3 Correlation. When performing the file test prescribed by 4.3.4.2, the following information shall be obtained.

a. The type and model of the filing machine used.

b. The type of file (roughness) used.

c. The length of the file stroke.

d. The pressure applied to the file.

e. The number of strokes of the file required to expose the basic metal.

The information thus obtained shall be the basis for determining compliance of the production sample hereinafter prescribed, with the requirements of paragraph 8.4.2 MIL-C-60536.

4.3.4.4 Deletion of the file test. The contractor, at his option, may select the wheel abrasive test (instead of the file test) to be used duing regular production. In this case paragraphs 4.3.4.2 and 4.3.4.3 do not apply.

4.4 Quality conformance inspection.

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4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation of MIL-A-48078.

4.4.1.1 Heat treatment lots. Heat treatment lots shall contain body forgings from not more than one heat of steel that has been heat treated in the same heat treatment equipment, at the same temperatures, in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Stress relieved lots. Stress relieved lots shall contain cold worked bodies from not more than one heat of steel that has been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, or stress relieved lots as the case may be. Such lots shall be identified by miscellaneous lot code numbers and processed in accordance with 4.4.1.1 or 4.4.1.2 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.4 Projectile lots. When it has been determined that bodies of heat treated, stress relieved or miscellaneous lots have complied with the mechanical properties requirements of the applicable drawing, they may be regrouped for assembly into projectile lots, to which will be assigned a projectile lot number. The projectile lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or work week. The projectile lot thus constituted shall be the basis for the selection of samples for the functional test hereinafter prescribed.

4.4.2 Examination. See MIL-A-48078.

a. Unless otherwise specified in the classification of defects and tests, sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Level II.

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b. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination.

c. Tool control dimensions. Dimensions marked tool control shall be gaged during first article testing and after each 250 items produced on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action shall immediately be instituted by the contractor. All material produced prior to periodic inspection shall be rejected and screened whenever a defective tool control dimension is found.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cap, Nose				10523477
					NEXT HIGHER ASSEMBLY 9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Pitch diameter of thread, max.		0.40%	3.1	Gage
102	Minor diameter of thread, max.		0 40%	3.1	Gage
103	Small outside diameter		0.40%	3.1	Gage
104	Flange diameter		0.40%	3.1	Gage
105	Length of full form thread, min.		0.40%	3.1	Gage
106	Leading edge radius		0.40%	3.1	Visual (See 6.2)
107	Metal defective		0.40%	3.8.2	Visual
108	Runout of small outside diameter with pitch diameter of threads and rear face (if applicable)		0.40%	3.1	Gage
109	Runout of flange with pitch diameter of thread and rear face (if applicable)		0.40%	3.1	Gage
MINOR					
201	Total length		0.65%	3.1	Gage
202	Thickness of flange		0.65%	3.1	Gage
203	Depth of cavity		0.65%	3.1	Gage
204	Improper surface finish		0.65%	3.1	Visual (See 6.2)
205	Foreign matter		0.65%	3.8.5	Visual
206	Burr		0.65%	3.8.4	Visual
207	Pitch diameter of thread, min.		0.65%	3.1	Gage
208	Thread damaged or malformed		0.65%	3.8.3	Visual (See 6.2)

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Collar Ring				9335527
					NEXT HIGHER ASSEMBLY
					9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Outside diameter		0.40%	3.1	Gage
102	Runout of outside diameter with inside diameter (if applicable)		0.40%	3.1	Gage
103	Thickness of ring		0.40%	3.1	Gage
104	Parallelism of front face with rear face		0.40%	3.1	Gage
105	Straightness of front face		0.40%	3.1	Gage
106	Straightness of rear face		0.40%	3.1	Gage
MINOR					
201	Metal defective		0.65%	3.8.2	Visual
202	Burr		0.65%	3.8.4	Visual
203	Foreign matter		0.65%	3.8.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.3	Spike				10523476
					NEXT HIGHER ASSEMBLY
					9331316
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Pitch diameter of cap thread, min.		0.40%	3.1	Gage
102	Major diameter of cap thread, min.		0.40%	3.1	Gage
103	Pitch diameter of body thread, min.		0.40%	3.1	Gage
104	Major diameter of body thread, min.		0.40%	3.1	Gage
105	Diameter of shoulder		0.40%	3.1	Gage
106	Diameter of stem (if applicable)		0.40%	3.1	Gage
107	Length to forward basic diameter (if applicable)		0.40%	3.1	Gage
108	Length to rear basic diameter (if applicable)		0.40%	3.1	Gage
109	Radius at juncture of spike stem and shoulder		0.40%	3.1	Visual (See 6.2)
110	Runout of front face of shoulder with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
111	Runout of forward basic diameter with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
112	Runout of shoulder diameter with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
NOTES:					

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MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 10523476
4.4.2.3	Spike				NEXT HIGHER ASSEMBLY 9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Runout of pitch diameter of cap thread with pitch diameter of body threads, body seat surface and cap seat surface (if applicable)		0.40%	3.1	Gage
114	Tension test		4.4.3.1.2	3.1	4.5.1.1.4
MINOR					
201	Diameter of cap thread undercut, min.		0.65%	3.1	Gage
202	Diameter of body thread undercut, min.		0.65%	3.1	Gage
203	Diameter of body thread pilot, min.		0.65%	3.1	Gage
204	Distance from rear to rear face of shoulder		0.65%	3.1	Gage
205	Distance from rear to front face of shoulder		0.65%	3.1	Gage
206	Distance from front face to cap seat		0.65%	3.1	Gage
207	Distance from cap seat to rear face		0.65%	3.1	Gage
208	Length of body thread pilot, max.		0.65%	3.1	Gage
209	Length of cap thread undercut, max.		0.65%	3.1	Gage
210	Length of body thread undercut, max.		0.65%	3.1	Gage
211	Radii or chamfers, missing or incorrect (except as otherwise classified)		0.65%	3.1	Visual (See 6.2)
212	Improper surface finish		0.65%	3.1	Visual (See 6.2)
213	Presence of driving slots (if applicable)		0.65%	3.1	Visual (See 6.2)

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 10523476
4.4.2.3	Spike				NEXT HIGHER ASSEMBLY 9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
214	Metal defective		0.65%	3.8.2	Visual
215	Foreign matter		0.65%	3.8.5	Visual
216	Burr		0.65%	3.8.4	Visual
217	Pitch diameter of body thread, max.		0.65%	3.1	Gage
218	Pitch diameter of cap thread, max.		0.65%	3.1	Gage
219	Threads damaged or malformed		0.65%	3.8.3	Visual (See 6.2)

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.4	Spike (For Swaged Nose Collar)		9331315
			NEXT HIGHER ASSEMBLY
			9331316

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Major diameter of body thread, min.		0.40%	3.1	Gage
102	Pitch diameter of body thread, min.		0.40%	3.1	Gage
103	Diameter of spike stem		0.40%	3.1	Gage
104	Diameter of front of spike stem		0.40%	3.1	Gage
105	Diameter of nose collar seat		0.40%	3.1	Gage
106	Diameter of shoulder		0.40%	3.1	Gage
107	Runout of nose collar seat diameter with pitch diameter of body thread and base seat surface		0.40%	3.1	Gage
108	Runout of front of spike stem diameter with pitch diameter of body threads and base seat surface		0.40%	3.1	Gage
109	Distance from rear of nose collar seat to front face of spike		0.40%	3.1	Gage
110	Distance from front of nose collar seat to front face of spike		0.40%	3.1	Gage
111	Radius at juncture of spike stem and shoulder		0.40%	3.1	Visual (See 6.2)
112	Runout of diameter of spike stem with pitch diameter of body threads and base seat surface (if applicable)		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.4	Spike (For Swagged Nose Collar)				9331315
					NEXT HIGHER ASSEMBLY 9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Runout of shoulder diameter with pitch diameter of body theads and base seat surface (if applicable)		0.40%	3.1	Gage
114	Perpendicularity of front face of shoulder with pitch diameter of body threads and base seat surface (if applicable)		0.40%	3.1	Gage
115	Tension test	4	4.3.1.2	3.1	4.5.1.1.4
MINOR					
201	Diameter of body thread undercut, min.		0.65%	3.1	Gage
202	Diameter of body thread pilot, min.		0.65%	3.1	Gage
203	Distance from rear to rear face of shoulder		0.65%	3.1	Gage
204	Distance from rear to front face of shoulder		0.65%	3.1	Gage
205	Length of body thread pilot, max.		0.65%	3.1	Gage
206	Length of body thread undercut, max.		0.65%	3.1	Gage
207	Radii or chamfers, missing or incorrect (except as otherwise classified)		0.65%	3.1	Visual (See 6.2)
208	Improper surface finish		0.65%	3.1	Visual (See 6.2)
209	Presence of driving slots (if applicable)		0.65%	3.1	Visual (See 6.2)
210	Metal defective		0.65%	3.8.2	Visual

NOTES:

DESC-DA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 9331315
4.4.2.4	Spike (For Swagged Nose Collar)				NEXT HIGHER ASSEMBLY 9331316
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
211	Foreign matter		0.65%	3.8.5	Visual
212	Burr		0.65%	3.8.4	Visual
213	Total length		0.65%	3.1	Gage
214	Pitch diameter of body thread, max.		0.65%	3.1	Gage
215	Threads damaged or malformed		0.65%	3.8.3	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9331316
4.4.2.5	Spike Assembly (With Nose Cap)				NEXT HIGHER ASSEMBLY 10523473
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Nose cap fails to withstand minimum assembly torque		0.40%	3.7	4.5.2.3
102	Runout of small diameter of nose cap with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
103	Runout of nose cap flange diameter with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
MINOR					
201	Protective coating missing or inadequate (if applicable)		0.65%	3.3	Visual
202	Burr		0.65%	3.8.4	Visual
203	Foreign matter		0.65%	3.8.5	Visual

NOTES:

DSMC-0A (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Spike Assembly (With Swaged Nose Collar)	1 OF 1	9331316
			NEXT HIGHER ASSEMBLY: 10523473

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Distance from front face of collar ring to front face of lip, min.		0.40%	3.1	Gage
102	Outside diameter of lip		0.40%	3.1	Gage
103	Runout of rear face of collar ring with pitch diameter of body threads and body seat surface		0.40%	3.1	Gage
104	Runout of outside diameter of collar ring with pitch diameter of body threads and body seat surface (if applicable)		0.40%	3.1	Gage
MINOR					
201	Protective coating missing or inadequate (if applicable)		0.65%	3.3	Visual
202	Burr		0.65%	3.8.4	Visual
203	Foreign matter		0.65%	3.8.5	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Fin				8597610
					NEXT HIGHER ASSEMBLY
					9327365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Minor diameter of base thread, max.		100%	3.1	Gage
102	Pitch diameter of base thread, max.		0.40%	3.1	Gage
103	Minor diameter of tracer thread, max.		0.40%	3.1	Gage
104	Pitch diameter of tracer thread, max.		0.40%	3.1	Gage
105	Length of effective base thread, min.		0.40%	3.1	Gage
106	Length of effective tracer thread, min.		0.40%	3.1	Gage
107	Position of tracer cavity with minor diameter of tracer thread		0.40%	3.1	Gage
108	Position of fin hub with minor diameter of tracer thread		0.40%	3.1	Gage
109	Position of fin hub with pitch diameter of base thread and the front face		0.40%	3.1	Gage
110	Total length		0.40%	3.1	Gage
111	Straightness of fin blades		0.40%	3.1	Gage
112	Diameter of fin blades		0.40%	3.1	Gage
113	Thickness of fin blade web		0.40%	3.1	Gage
114	Angle at front of fin		0.40%	3.1	Gage
115	Diameter of fin hub		0.40%	3.1	Gage
116	Threads damaged or malformed		0.40%	3.8.3	Visual (See 6.2)
117	Metal defective		0.40%	3.8.2	Visual (See 6.2)
118	Tension test	4.	4.3.1.2	3.1	4.5.1.1.2

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.7	Fin		8597610
			NEXT HIGHER ASSEMBLY
			9327365

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Abrasion resistance test (See Note 1)	4	4.3.2	3.3	4.5.3.2
120	Thickness of protective coating (See Note 1)		0.40%	3.3	Gage
121	Width of fin blades (T Section)		0.40%	3.1	Gage
122	Angle between fin blades		0.40%	3.1	Gage
MINOR					
201	Pitch diameter of tracer thread, min.		0.65%	3.1	Gage
202	Diameter of tracer thread undercut, max.		0.65%	3.1	Gage
203	Diameter of tracer cavity		0.65%	3.1	Gage
204	Depth of tracer thread cavity		0.65%	3.1	Gage
205	Depth from rear face to bottom of tracer thread cavity		0.65%	3.1	Gage
206	Depth of base thread cavity		0.65%	3.1	Gage
207	Radii or chamfers missing or incorrect		0.65%	3.1	Visual (See 6.2)
208	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
209	Foreign matter		0.65%	3.8.5	Visual
210	Burr		0.65%	3.8.4	Visual
211	Protective coating on front face excessive		0.65%	3.1	Visual
212	Diameter of base thread counterbore, max.		0.65%	3.1	Gage
213	Pitch diameter of base thread, min.		0.65%	3.1	Gage

NOTES: 1. This test can be performed in the fin and base assembly inspection (para. 4.4.2.9).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.4.2.8	Base				10523474
					NEXT HIGHER ASSEMBLY
					9327365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Pitch diameter of body thread, min.		0.40%	3.1	Gage
102	Major diameter of body thread, min.		100%	3.1	Gage
103	Pitch diameter of fin thread, min.		0.40%	3.1	Gage
104	Major diameter of fin thread, min.		100%	3.1	Gage
105	Outside diameter of front straight section		0.40%	3.1	Gage
106	Diameter of rear straight section		0.40%	3.1	Gage
107	Depth of crimp groove		0.40%	3.1	Gage
108	Distance from body seat surface to outside basic diameter		0.40%	3.1	Gage
109	Length of effective fin thread, min.		0.40%	3.1	Gage
110	Location of crimp groove		0.40%	3.1	Gage
111	Profile of radius at rear of crimp grooves		0.40%	3.1	Gage
112	Profile of taper		0.40%	3.1	Gage
113	Protective coating on threads		0.40%	3.3	Visual
114	Metal defective		0.40%	3.8.2	Visual
115	Threads damaged or malformed		0.40%	3.8.3	Visual (See 6.2)
116	Tension test	4.4.3.1.2		3.1	4.5.1.1.3
117	Abrasion resistance test (See Note 1)	4.4.3.2		3.3	4.5.3.2
118	Thickness of protective coating (See Note 1)		0.40%	3.3	Gage

NOTES:
1. This test can be performed in the fin and base assembly inspection (para. 4.4.2.9).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
4.4.2.8	Base				10523474
					NEXT HIGHER ASSEMBLY
					9327365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
119	Runout of pitch diameter of fin thread with pitch diameter of body thread, fin seat surface and body seat surface (if applicable)		0.40%	3.1	Gage
120	Runout of rear straight section with pitch diameter of body thread and body seat surface (if applicable)		0.40%	3.1	Gage
121	Runout of front straight section with pitch diameter of body thread and body seat surface (if applicable)		0.40%	3.1	Gage
MINOR					
201	Diameter of body thread undercut, min.		0.65%	3.1	Gage
202	Diameter of fin thread pilot		0.65%	3.1	Gage
203	Diameter of cavity, max.		0.65%	3.1	Gage
204	Depth of cavity, max.		0.65%	3.1	Gage
205	Distance from front face to body seat surface		0.65%	3.1	Gage
206	Distance from rear face to fin seat surface		0.65%	3.1	Gage
207	Length of fin thread pilot		0.65%	3.1	Gage
208	Total length		0.65%	3.1	Gage
209	Length of body thread undercut, max.		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.8	Base				10523474
					NEXT HIGHER ASSEMBLY
					9327365
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
210	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
211	Presence of driving slots		0.65%	3.1	Visual (See 6.2)
212	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
213	Foreign matter		0.65%	3.8.5	Visual
214	Burr		0.65%	3.8.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.9	Fin and Base Assembly				9327365
					NEXT HIGHER ASSEMBLY
					10523473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1.	Gap between fin and base		100%	3.1	Gage
MAJOR					
101	Distance from body seat surface to rear face, max.		0.40%	3.1	Gage
102	Torque test		0.40%	3.7	4.5.2.1
103	Runout of fins with pitch diameter of body thread and body seat surface (if applicable)		0.40%	3.1	Gage
104	Angle between fin blades (See dwg. 8597610)		0.40%	3.1	Gage
105	Abrasion resistance test (if applicable)		4.4.3.2	3.3	4.5.3.2
106	Thickness of protective coating (if applicable)		0.40%	3.3	Gage
107	Metal defective		0.40%	3.8.2	Visual (See 6.2)
108	Threads damaged or malformed		0.40%	3.8.3	Visual (See 6.2)
109	Protective coating inadequate or damaged		0.40%	3.3	Visual
110	Presence of sealant		0.40%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9327365
4.4.2.9	Fin and Base Assembly				NEXT HIGHER ASSEMBLY 10523473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Protective coating on body seat surface, excessive		0.65%	3.1	Visual
202	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
203	Surface finish improper		0.65%	3.1	Visual (See 6.2)
204	Foreign matter		0.65%	3.8.5	Visual
205	Burr		0.65%	3.8.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.10	Obturator				9327293
					NEXT HIGHER ASSEMBLY
					10523473
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Total length		0.40%	3.1	Gage
103	Location of the basic diameter		0.40%	3.1	Gage
104	Runout of outside diameter to inside diameter		0.40%	3.1	Gage
105	Ultimate Load				
	(a) 140°F	Note 1	2.5%	3.1	4.5.5
	(b) 73°F	Note 1	2.5%	3.1	4.5.5
	(c) -35°F	Note 1	2.5%	3.1	4.5.5
106	Extension				
	(a) 140°F	Note 1	2.5%	3.1	4.5.5
	(b) 73°F	Note 1	2.5%	3.1	4.5.5
	(c) -35°F	Note 1	2.5%	3.1	4.5.5
107	Hardness	Note 1	2.5%	3.1	4.5.7
108	Specific gravity	3		3.1	4.5.8
109	Moisture content and water absorption	5		3.1	4.5.9
110	Arrows missing, illegible or in the wrong direction		0.40%	3.1	Visual
111	Evidence of poor workmanship		0.40%	3.8	Visual
112	Molding defects		0.40%	3.8.6	Visual
113	Profile of external taper		0.40%	3.1	Gage
114	Chamfer incorrect		0.40%	3.1	Gage

NOTES: 1. A special inspection level S-3 with a double sampling plan according to MIL-STD-105 shall be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.10	Obturator				9327293
					NEXT HIGHER ASSEMBLY
					10523473
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
MINOR					
201	Surface finish, improper		0.65%	3.1	Visual (See 6.2)
202	Excessive flash		0.65%	3.8.1	Visual
203	Foreign matter		0.65%	3.8.5	Visual
204	Burr		0.65%	3.8.4	Visual
NOTES:					

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Seal				9327359
					NEXT HIGHER ASSEMBLY
					9327295
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Total length		0.40%	3.1	Gage
103	Inside chamfer		0.40%	3.1	Gage
104	Tensile				
	(a) 140°F	Note 1	2.5%	3.1	4.5.6
	(b) 73°F	Note 1	2.5%	3.1	4.5.6
	(c) -35°F	Note 1	2.5%	3.1	4.5.6
105	Extension				
	(a) 140°F	Note 1	2.5%	3.1	4.5.6
	(b) 73°F	Note 1	2.5%	3.1	4.5.6
	(c) -35°F	Note 1	2.5%	3.1	4.5.6
106	Specific gravity	3		3.1	4.5.8
107	Moisture content and water absorption	5		3.1	4.5.9
108	Evidence of poor workmanship		0.40%	3.8	Visual
109	Surface finish, improper		0.40%	3.1	Visual (See 6.2)
110	Molding defects		0.40%	3.8.6	Visual
111	Quality of the etching bath	4.4.3.4		3.1	4.5.2.2
112	Quality of the mechanical abrasion	4.4.3.5		3.1	4.4.3.5
MINOR	None defined				

NOTES:
1. A special inspection level S-3 with a double sampling plan according to MIL-STD-105 shall be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.12	Body				10523475
					NEXT HIGHER ASSEMBLY
					9327295
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Runout of inside diameter with bourrelets		0.40%	3.1	Gage
102	Runout of body to the bourrelets		0.40%	3.1	Gage
103	Pitch diameter of spike thread, max.		0.40%	3.1	Gage
104	Minor diameter of spike thread, max.		0.40%	3.1	Gage
105	Pitch diameter of base thread, max.		0.40%	3.1	Gage
106	Minor diameter of base thread, max.		0.40%	3.1	Gage
107	Diameter of bourrelets, min.		0.40%	3.1	Gage
108	Inside diameter of body (cavity)		0.40%	3.1	Gage
109	Outside diameter of body (center section)		0.40%	3.1	Gage
110	Diameter of obturating band seat		0.40%	3.1	Gage
111	Length of full form of spike thread, min.		0.40%	3.1	Gage
112	Length of obturating band seat		0.40%	3.1	Gage
113	Length to basic diameter		0.40%	3.1	Gage
114	Metal defective		0.40%	3.8.2	Visual (See 6.2)
115	Threads damaged or malformed		0.40%	3.1	Visual (See 6.2)
116	Tension test	4.4.3.1.1		3.1	4.5.1.1.1
117	Runout of seal seat surface with bourrelets (if applicable)		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 10523475
4.4.2.12	Body				NEXT HIGHER ASSEMBLY 9327295
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
118	Runout of pitch diameter of base thread with bourrelets and rear face (if applicable)		0.40%	3.1	Gage
119	Runout of pitch diameter of spike thread with bourrelets and front face (if applicable)		0.40%	3.1	Gage
MINOR					
201	Depth of spike thread cavity		0.65%	3.1	Gage
202	Length of front bourrelet		0.65%	3.1	Gage
203	Length of rear bourrelet		0.65%	3.1	Gage
204	Total length		0.65%	3.1	Gage
205	Radii or chamfers, missing or incorrect		0.65%	3.1	Visual (See 6.2)
206	Improper surface finish		0.65%	3.1	Visual (See 6.2)
207	Foreign matter		0.65%	3.8.5	Visual
208	Burr		0.65%	3.8.4	Visual
209	Pitch diamter of spike thread, min.		0.65%	3.1	Gage
210	Pitch diameter of base thread, min.		0.65%	3.1	Gage
211	Protective coating missing or inadequate		0.65%	3.3	Visual
212	Presence of excessive coating on seal seat surface		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9327295
4.4.2.13	Body and Seal Assembly				NEXT HIGHER ASSEMBLY 10523473
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Major diameter of seal		0.40%	3.1	Gage
102	Distance from base seat surface to rear face of seal flange		0.40%	3.1	Gage
103	Presence of gap at seal/body joint		0.40%	3.1	Gage
104	Runout of major diameter of seal with bourrelets		0.40%	3.1	Gage
105	Seal above flush with base seat		0.40%	3.1	Gage
106	Sealant not present for full 360°		0.40%	3.1	Visual
107	Sealant build-up on surfaces		0.40%	3.1	Visual
108	Torque test	Note 1		3.1	4.5.2.2
109	Seal above flush with body bourrelet		0.40%	3.1	Gage
MINOR					
201	Body seal area not properly cleaned		0.65%	3.1	Visual
202	Body protective coating damaged		0.65%	3.1	Visual
203	Surface finish improper		0.65%	3.1	Visual (See 6.2)
204	Radius incorrect		0.65%	3.1	Gage
205	Seal below flush with base seat, max.		0.65%	3.1	Gage
206	Seal below flush with body bourrelet, max.		0.65%	3.1	Gage
207	Foreign matter		0.65%	3.8.5	Visual

NOTES:
1. One body and seal assembly shall be tested at the middle and end of each shift at each station.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.14	Metal Parts Assembly (Before Painting)				10523473
					NEXT HIGHER ASSEMBLY
					9331331
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Spike fails to withstand minimum assembly torque		0.40%	3.7	4.5.2.3
102	Runout of small diameter of nose cap with bourrelets (if applicable)		0.40%	3.1	Gage
103	Runout of diameter of nose cap flange with bourrelets (if applicable)		0.40%	3.1	Gage
104	Runout of diameter of spike stem with bourrelets (if applicable)		0.40%	3.1	Gage
105	Runout of diameter of spike shoulder with bourrelets		0.40%	3.1	Gage
106	Perpendicularity of front face of spike shoulder with bourrelets		0.40%	3.1	Gage
107	Fin and base assembly fails to withstand minimum assembly torque		0.40%	3.7	4.5.2.3
108	Runout of base at rear of crimp groove with bourrelets		0.40%	3.1	Gage
109	Runout of small diameter of base with bourrelets		0.40%	3.1	Gage
110	Runout of fin with bourrelets		0.40%	3.1	Gage
111	Gap between base and body		0.40%	3.1	Gage
112	Gap between spike and body		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.14	Metal Parts Assembly (Before Painting)				10523473
					NEXT HIGHER ASSEMBLY
					9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Obturator inverted (arrows direction)		0.40%	3.1	Visual
114	Protective coating of fin and base missing or damaged		0.40%	3.3	Visual
115	Runout of outside diameter of collar ring with bourrelets (if applicable)		0.40%	3.1	Gage
116	Runout of front of spike stem (spike assembly with swagged nose collar) with bourrelets (if applicable)		0.40%	3.1	Gage
MINOR					
201	Burr		0.65%	3.8.4	Visual
202	Foreign matter		0.65%	3.8.5	Visual
NOTES:					

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-60811B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Metal Parts Assembly (After Painting)				10523473
					NEXT HIGHER ASSEMBLY 9331331
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR					
101	Weight incorrect		0.40%	3.1	Approved Scale
102	Diameter of bourrelets, max.		0.40%	3.1	Gage
103	Length, max.		0.40%	3.1	Gage
104	Target Impact Dispersion ballistic test	10		3.5.2	4.5.4.2
105	Excess pressure test	10		3.5.1	4.5.4.1
106	Salt spray test	See Note 1		3.3	See Note 1
107	Adhesion of paint	See Note 1		3.3	See Note 1
MINOR					
201	Protective coating inadequate or missing		0.65%	3.3	Visual
202	Marking missing, incorrect or illegible		0.65%	3.1	Visual
203	Burr		0.65%	3.8.4	Visual
204	Foreign matter		0.65%	3.8.5	Visual

NOTES: 1. The salt spray test (use ASTM B117 to perform this test) and the adhesion of paint test shall be conducted in accordance with TT-C-490. The actual projectile assembly shall be used in these tests. The sampling plan and the rejection criteria is defined in TT-C-490.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Carton, Packed (prior to sealing)				NA
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Support missing or damaged		0.65%	3.1	Visual
202	Separator missing or damaged		0.65%	3.1	Visual
203	Improper assembly		0.65%	3.1	Visual

NOTES:

DPSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Carton, Packed (after sealing)				NA
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR			0.65%	3.1	Manual
201	Contents loose		0.65%	3.1	Visual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not completely sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible				

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Pallet				7548604
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined				
MAJOR	None defined				
MINOR					
201	Strapping missing, loose or broken		0.65%	3.1	Visual and Manual
202	Pallet damaged (unserviceable)		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Marking missing, incorrect, or illegible		0.65%	3.1	Visual

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.3.1 Tension test

4.4.3.1.1 Body. Two bodies from each heat treatment, stress relieved, or miscellaneous lot as defined by 4.4.1.1 through 4.4.1.3 as applicable, after final heat treatment or stress relief shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any tension test specimen to comply shall be cause for rejection of the lot of bodies represented by the failing specimen.

4.4.3.1.2 Finished aluminum components. Two spikes, two bases, and two fins from each 1200 of each component or fraction thereof shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot of components represented by the failing sample. At the discretion of the procuring activity this test may be performed on the raw material, using one test specimen for each 1000 pounds of material or fraction thereof, except that the test performed by the Government shall be made on specimens taken from the finished components.

4.4.3.2 Abrasion resistance test

4.4.3.2.1 Fin and base assembly - file abrasive test. One fin and base assembly (or set of components thereof) from each eight hours production or fraction thereof from the protective coating process shall be subjected to this test to determine compliance with paragraph 9.4.4 of MIL-C-60536, except that the information obtained during the file test in 4.3.4.2 shall be utilized as indicated in 4.3.4.3. Failure of the assembly or components to comply with the applicable requirements shall be cause for rejection of the product processed since the last acceptable test was performed.

4.4.3.2.2 Test panels - wheel abrasive test. If the contractor decides to use the wheel abrasive test during regular production (see 4.3.4.4), two test panels as specified in paragraph 9.4.3 of MIL-C-60536 shall be processed along with the product from each eight hours production. These test panels shall be subjected to a test in accordance with paragraph 9.4.3 of MIL-C-60536. Failure of any test panel to comply with the applicable requirement shall be cause for rejection of the product processed since the last acceptable test was performed.

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4.4.3.3 Functional. The following proving ground tests will be performed by the Government:

4.4.3.3.1 First article sample. The quantities listed in Table II shall be ballistically tested in the phases indicated.

TABLE II

Phase	Temp.	Tube Wear	Qty.	Requirement
Target-Impact-Dispersion	70 ± 2.5°F (21 ± 1.5°C)	New	25	Target-impact-dispersion & metal part security. Muzzle velocity, pressure, obturator discard & spin rate for information only.
Excess Pressure	70 ± 2.5°F (21 ± 1.5°C)	New	25	Metal parts security. All others for information only.

4.4.3.3.2 Production lot sample. The quantities listed in Table III shall be ballistically tested in the phases indicated.

TABLE III

Phase	Temp.	Tube Wear	Qty.	Requirement
Target-Impact-Dispersion	70 ± 2.5°F (21 ± 1.5°C)	New	10	Target-impact-dispersion & metal part security. Muzzle velocity, pressure, obturator discard for information only.
Excess Pressure	70 ± 2.5°F (21 ± 1.5°C)	New	10	Metal parts security. All others for information only.

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4.4.3.3.3 Security. Failure of any projectile assembly to comply with 3.5.1 shall be cause for rejection of the first article sample or the production lot during the performance of any ballistic test.

4.4.3.3.4 Target-impact-dispersion. Samples provided in accordance with Table II and III will be subjected to this test. If the pooled Probable Error (PE) of the sample, calculated in accordance with 6.5, multiplied by the appropriate factor of Table IV (see 6.11) exceeds the requirements of paragraph 3.5.2, the lot shall be rejected.

TABLE IV

Sample Size	Degrees of Freedom	Factor
25 - Test Rounds	20	.80
24 - (one target miss)	19	.80
15 - Calibration Rounds	12	.76
10 - Test or Calibration Rounds	8	.72
9 - (one target miss)	7	.71

4.4.3.4 Adhesion testing. The quality of the etching bath shall be monitored by assembling a seal, etched in the bath that is to be tested, to a body and torque testing the assembly. The sample size shall be two assemblies per occasion. The etch solution shall be tested before and after each change to the bath and no less than once very eight hours of production. The samples shall be tested in accordance with paragraph 4.5.2.2. Failure of any sample to comply with the minimum torque requirement of the applicable drawing shall be cause for rejection of all seals etched with that bath or etched subsequent to the last test, whichever is applicable (see 6.10).

4.4.3.5 Mechanical abrasion (if applicable). The quality of the mechanical abrasion of the seals shall be monitored by taking five (5) seals at the middle and end of each shift and inspecting for the following characteristics:

a. Presence of surface sheen

b. Increase in inside diameter

c. Average surface finish increase (to be measured in at least 3 places, 120° apart.

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Failure of any sample to comply with the requirements of the applicable drawing shall be cause for rejection of all seals mechanically abraded since the last acceptable test.

4.4.3.6 Seal assembly test. At the middle and end of each shift one body and seal assembly per station shall be tested in accordance with paaragraph 4.5.2.2. Failure of any sample shall be cause for rejection of all assemblies assembled since the last acceptable test.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment design in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.7 herein.

4.5 Test method and procedures

4.5.1 Mechanical properties. The minimum mechanical properties requirements shall apply throughout the component unless otherwise designated. The number of specimens and their location are for the convenience of the contractor and government and shall not restrain the contractor or government from testing additional specimens at other locations to determine that properties of the components conform to drawing requirements.

4.5.1.1 Tension test

4.5.1.1.1 Body. From the bodies provided in accordance with 4.4.3.1.1, two tension test specimens conforming to the largest obtainable standard round specimens, as prescribed by ASTM E8, shall be taken longitudinally from opposite sides of the wall of each sample between the forward and rear bourrelets or from any other location deemed necessary by the government. The criteria for determining yield strength by the offset method shall be as prescribed by ASTM E8. The limiting set shall be 0.1 percent (0.001 inch per inch of gage length).

4.5.1.1.2 Fin. From each sample selected in accordance with 4.4.3.1.2, one tension test specimen conforming to the largest obtainable round specimen prescribed by ASTM E8 shall be selected. The location of the specimen shall be at the convenience of the contractor. The criteria for performing this test shall be as prescribed by ASTM E8. The limiting set shall be 0.2 percent.

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4.5.1.1.3 Base. From each sample supplied in accordance with 4.4.3.1.2 for this component, one tension test specimen conforming to the largest obtainable round specimen as prescribed by ASTM E8, shall be taken longitudinally from each sample. The specimen shall be tested using the criteria for performing this test as prescribed by ASTM E8. The limiting set shall be 0.2 percent.

4.5.1.1.4 Spike. From each sample supplied in accordance with 4.4.3.1.2 for this component, one tension test specimen conforming to the largest obtainable round specimen as prescribed by ASTM E8, shall be taken from the small diameter of each sample. The sample shall be tested using the criteria for performing this test as prescribed by ASTM E8. The limiting set shall be 0.2 percent.

4.5.2 Torque test

4.5.2.1 Fin and base assembly. The fin and base assemblies shall be mounted in an approved holding device in such a manner that the base is prevented from turning. The required disassembly torque shall be applied to the fin gradually in the direction required for disassembly. Any perceptible movement of the fin shall be cause for rejection of the fin and base assembly. Testing shall be conducted after thread sealant is fully cured.

4.5.2.2 Body and seal assembly. The body and seal assembly shall be mounted in such a manner that the body is prevented from moving. A line is then scribed across both the body and seal in such a manner that movement can be easily observed. The torque fixture shall be placed over the unmachined seal and radial holes drilled into the seal for the four holding pins. Holes shall be drilled to a depth to prevent seal tearing but not into the body material. The holding pins shall be inserted into the holes and a static torque applied to the seal. No torque shall be applied to the seal until the adhesive is fully cured. The test fixture shall be in accordance with Figure II or its equivalent.

4.5.2.3 Body and spike assembly, body and fin-base assembly, spike and nose cap assembly. In performing these tests, a torque equal to that prescribed for assembly by the applicable drawing shall be applied in the direction of assembly while the thread sealant is still wet. The joint under test shall be marked to provide a means for detection of any movement between the adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.5.3 Protective coating

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4.5.3.1 Body and spike. The procedures for testing the cleaning of the components and the application of protective coating shall be in accordance with the criteria for testing as specified in the applicable specification.

4.5.3.2 Abrasion resistance. The sample provided by 4.4.3.2.1 shall be tested using a filing machine of a type and model that was used in establishing correlation as indicated in 4.3.4.2. The samples shall be tested using the same type (roughness) file, the same length of stroke, the same number of strokes and the same file pressure as was used on the first article sample. If the contractor decides to do the wheel abrasive test (4.4.3.2.2), the test must be performed in accordance with paragraph 9.4.3 of MIL-C-60536.

4.5.4 Functional tests. All ballistic tests shall be conducted at a Government proving ground in accordance with the applicable approved Acceptance Test Procedures. The sample cartridges shall be fired from a M60 tank series and observations made for conformance to all applicable requirements. Photographic equipment will be set up near the muzzle in such a manner as to record security of projectile metal parts. When ballistically testing the projectile metal parts, reference/calibration components shall be used as a minimum in the target-impact-dispersion phase.

4.5.4.1 Excess pressure test. The metal parts assemblies supplied for the excess pressure phase, in accordance with 4.4.3.3.1 and 4.4.3.3.2 will be assembled into complete rounds and fired. The propellant charge will be adjusted to obtain a chamber pressure of 70,000 psi (482.6 MPa) to 73,900 psi (509.4 MPa).

4.5.4.2 Target-Impact-Dispersion test.

4.5.4.2.1 Firing sequence. The metal parts assemblies supplied in accordance with this phase of 4.4.3.3.4, assembled into complete rounds, will be fired according to the applicable Acceptance Test Procedure utilizing the service charge. The data generated in the new tube phase will primarily be used for determination of acceptance from a target-impact-dispersion standpoint, and will require the firing of reference/calibration rounds to establish a velocity correction. The firing sequence for the first article sample will be five (5) calibration rounds, fifteen (15) test rounds, five (5) calibration rounds, ten (10) test rounds, five (5) calibration rounds. For the production lot sample the firing sequence will be five (5) calibration rounds, ten (10) test rounds, five (5) calibration rounds. Velocity, spin

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rate and pressure data shall be recorded during the first article test. For the production lot tests, velocity data shall be recorded for each round fired (for info only, see 6.4). Also for the production lot tests, pressure data shall be collected during the first five lots, then every fifth lot thereafter, for information only.

4.5.4.2.2 Target. Target distance is to be 2000 meters and target size is to be 20-feet X 20-feet (6.1 meters X 6.1 meters) minimum.

4.5.4.2.3 Temperature conditioning. The ammunition shall be conditioned to the appropriate temperature until all propellant is stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours.

4.5.4.2.4 Aiming. After loading and locking a round in the chamber, the gun tube shall be traversed off target center, dropped down below target center, traversed to the opposite side of target center, elevated to target center elevation, and traversed onto target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above, or beyond target center do not reverse direction of gun tube movement, but repeat the sequence.

4.5.4.2.5 Spotting the target. Spotter rounds will be used to assure the weapon is zeroed on target center. A minimum of three consecutive target hits within one mil of target center are required. After the target has been spotted for a shot group, every effort should be made to keep the elevation and deflection of the weapon the same for that shot group.

4.5.4.2.6 Firing groups. The test samples shall be fired in five round groups (or occasions) with no less than three minutes but not more than five minutes between consecutive rounds in the same group. If more than a thirty minute lapse occurs between group firings, or if there was an indication of a trend in the previous group's target impacts to move away from the center of the target the weapon will be rezeroed on target center before firing the next group.

4.5.4.2.7 Target impact measurements. Target impacts (including spotters) shall be measured to the nearest one-half inch (1 centimeter) in both the horizontal and vertical distance from target center.

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4.5.4.2.8 Wind velocity. Wind velocity and gusts shall be measured at the firing site prior to the firing of each round in the target-dispersion phase ballistic test. The dispersion portion of the ballistic test shall not be fired if the surface wind velocity (nominal) exceeds 10 knots (18.5 Km/hr) and surface wind gusts exceeds 5 knots (9.25 Km/hr). In addition, peak surface wind velocity (nominal plus gusts) is not to exceed 10 knots (18.5 Km/hr) during the firing of any group during the dispersion phase.

4.5.4.3 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, the results shall be invalidated and additional cartridges, as required, shall be tested. A lot shall be retested whenever it fails to meet the target-impact-dispersion requirements as a result of being affected by the following conditions:

a. The wind velocity exceeds 10 knots (18.5 Km/hr) during the firing of any test group.

b. The cadence between any two consecutive rounds in a test group exceeds 5 minutes.

c. A change was made in the azimuth or elevation of the weapon within a test group.

d. Both the test samples and the calibration samples fail to meet the target-impact-dispersion requirements.

4.5.4.3.1 Target miss. Projectile lots which had one target miss during the target-impact-dispersion phase of the proving ground test described herein, shall be retested if the pooled PE of the remaining rounds multiplied by the appropriate factor (see 4.4.3.3.4) meets the requirements of 3.5.2. Two or more target misses by the test rounds shall reject the lot. Target miss coordinates will be estimated and recorded for all test phases, along with pertinent flight observations. Results of retests for target-impact-dispersion shall be referred to the contracting officer for disposition.

4.5.5 Obturator, extension and ultimate load. The ultimate load and extension shall be determined in accordance with ASTM D-2290 with the below listed exceptions. The test fixture shall be in accordance with Figure I or its equivalent. Machine loading shall be 0.5 inches per minute $\pm$ 0.1 inch per minute. Sample specimens each shall be tested at -35°F $\pm$ 4°F (-37°C $\pm$ 2°C), 73°F $\pm$ 4°F (22°C $\pm$ 2°C), and

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140°F ± 4°F (60°C ± 2°C). The ultimate load will be the maximum force recorded during testing and will be independent of cross-sectional area. For elongation the split disc shall be separated until each half is in contact with the specimen without applying a load to the specimen. This distance of separation (initial separation) shall then be measured. The specimen rupture force shall be measured. Split disc separation upon specimen rupture (ultimate separation) shall be measured. Extension shall be determined by subtracting initial separation from ultimate separation. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. The specimens used for this test shall be as received obturators.

4.5.6 Seal, extension and tensile. The ultimate tensile strength and extension shall be determined in accordance with ASTM D-2290. The test fixture shall be in accordance with Figure 1 or its equivalent. Sample specimens each shall be tested at -35°F ± 4°F (-37°C ± 2°C), 73°F ± 4°F (22°C ± 2°C), and 140°F± 4°F (60°C ± 2°C). The split disc shall be separated until each half is in contact with the specimen without applying a load to the specimen. The distance of separation (initial separation) shall then be measured. Split disc separation upon specimen rupture (ultimate separation) shall be measured. Extension shall be determined by subtracting initial separation from ultimate separation. Machine loading shall be two (2) inches per minute ± 0.5 inches per minute. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. The specimens used for this test shall be as received seals.

4.5.7 Obturator hardness. Rockwell hardness determination shall be made in accordance with ASTM D785, using the R scale with the exceptions that the hardness test shall be performed on the actual machined obturator and the specimens shall be conditioned for a minimum of 24 hours. Four (4) readings 90° apart shall be taken on the outside of the sample (radial direction). Failure of any reading to meet the applicable requirement shall be cause for rejection of the lot.

4.5.8 Seal and obturator, specific gravity. Three (3) seals and three (3) obturators shall be selected from each of the lots of seals and obturators for use in determining the specific gravity. The test shall be performed in accordance with method A1 per ASTM D792. Failure of any sample to meet the applicable requirements shall be cause for rejection of the lot.

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4.5.9 Seal and obturator, moisture content and water absorption. The bands shall be wiped clean with an OSHA acceptable solvent, i.e., dichlorotetraflourathane, force air dried, then weighed on an analytical balance. This is the initial weight. The bands are then placed in an oven maintained at 160°F ± 4°F (71°C ± 2°C) for twenty-four (24) hours. Cool in a desiccator and reweigh. Repeat oven conditioning and cooling until constant weight is achieved. This is the conditioned weight. Completely immerse the dried bands in a container of distilled water, maintained at 73°F ± 4°F (22°C ± 2°C) for a period of twenty-four (24) hours minus zero (0) plus one-half (1/2) hour. Bands shall be individually removed from the water, all surface water wiped off with a dry, lint-free cloth and weighed immediately on an analytical balance. This gives the wet weight. Percent of moisture content shall be calculated as follows:

$$\frac{\text{Initial weight - conditioned weight}}{\text{Initial weight}} \times 100 = \% \text{ moisture content}$$

Percent water absorption shall be calculated as follows:

$$\frac{\text{Wet weight - conditioned weight}}{\text{Conditioned weight}} \times 100 = \% \text{ water absorption}$$

Failure of any sample to meet the applicable requirement shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing

5.1.1 Level B minimum military pack. The metal parts for projectile, 105MM TP-T, M489 shall be packed as follows:

5.1.1.1 Metal parts assembly. Part No. 10523473 shall be packed in accordance with drawing 7548604.

5.1.2 Percentage of extra parts packing. Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.1.3 Shipping. If the component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

5.1.4 Marking. The pallet shall be marked in accordance with drawing 7548604. The carton shall be marked in accordance with MIL-STD-1169.

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6. NOTES

6.1 Intended use. The TP-T M489 projectile metal parts assembly is used in the TP-T M490 cartridge and M68 gun.

6.1.1 Ordering data. See MIL-A-48078.

6.2 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.3 In checking the thickness of protective coating on the fin and base the "Dermitron" by Unit Process Assemblies Inc. 53-15 37th Ave., Woodside, NY 11377, has been proven to be satisfactory.

6.4 Velocity. Proving ground tests for velocity are not acceptance tests, but the results will be carefully analyzed by US ARDC to determine quality of the ammunition produced.

6.5 Method of calculating pooled PE. The individual PE results of the groups (5 samples in each group) of samples shall be pooled in accordance with the following formula:

$$PE_{pooled} = \sqrt{\frac{\sum\left[(N_i - 1)\ PE_i^2\right]}{\sum (N_i - 1)}}$$

Where: i - Number of groups
N_i - Number of samples in a group (target hits)
PE_i - Probable Error of the group

6.6 New Tube wear definitions. A new tube shall not have the primary land wear exceed .015 inches. Primary wear is measured in a vertical plane 25.25 inches from the rear face of the tube.

6.7 Submission of inspection equipment designs for approval. See MIL-A-48078.

Commander
US Army Armament Research and Development Center
ATTN: DRSMC-QAT-I(D)
Dover, NJ 07801

6.8 Submission of test data and ammunition data cards. In addition to the normal distribution of records when the projectile is procured by the Army Armament, Munitions and Chemical Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following: Commander, ARDC, ATTN: DRSMC-LCU-M(D), DRSMC-LCU-CV(D) and DRSMC-QAT-A(D), Dover, NJ 07801.

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6.9 Sulfuric acid - dicromate solution. The sulfuric acid dicromate solution shall be prepared in accordance with the applicable drawing.

6.10 Seal rework. If any test sample fails to meet the requirement the rejected seals may be re-etched provided that the total etch time does not exceed the maximum allotted time.

6.11 Probable error factor. The factors shown in Table IV are taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A .05) and (N-1) degrees of freedom.

6.12 Probable error calculations. The probable error of the group is calculated in accordance with the following equation:

$$PE_{mils} = \frac{(.6745)\ (1000)\ (SD)}{(X)\ \ (Rg)}$$

Where:

SD - Standard deviation
Rg - Range in meters
X - Conversion factor
a) when target hits are measured in inches, X = 39.37
b) when target hits are measured in centimeters, X = 100.0

6.13 Standard deviation. The standard deviation of a sample is calculated in accordance with the following equation found on page 1-10 AMCP 706-110:

$$SD = \frac{N \quad X^2 \quad - \quad (\quad X)^2}{N(N-1)}$$

Where:

N - the number in the sample
X - the individual data point

6.14 Visual standards. Visual standards are to be established at the contractor's plant for compliance with 3.8.2.

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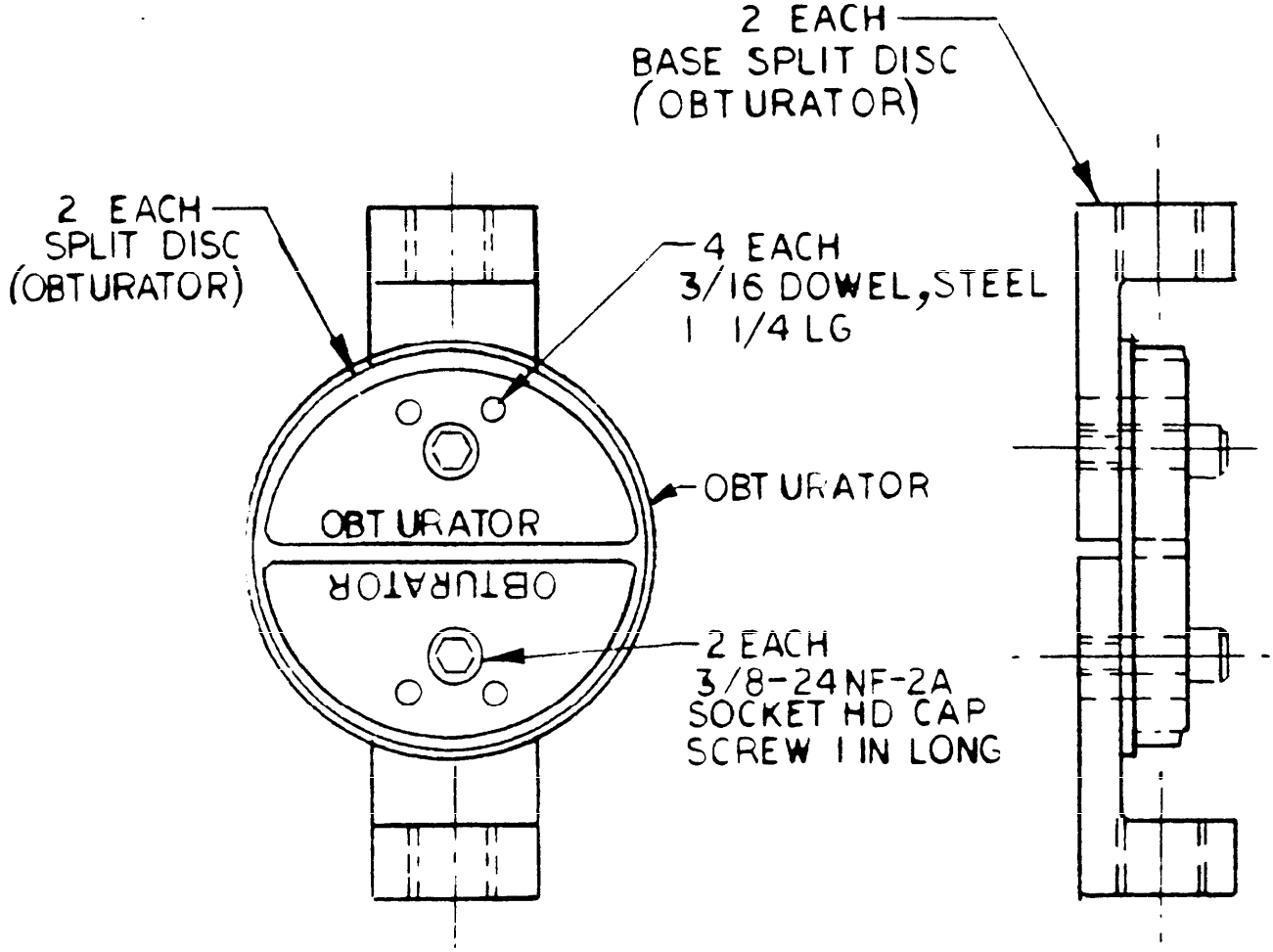
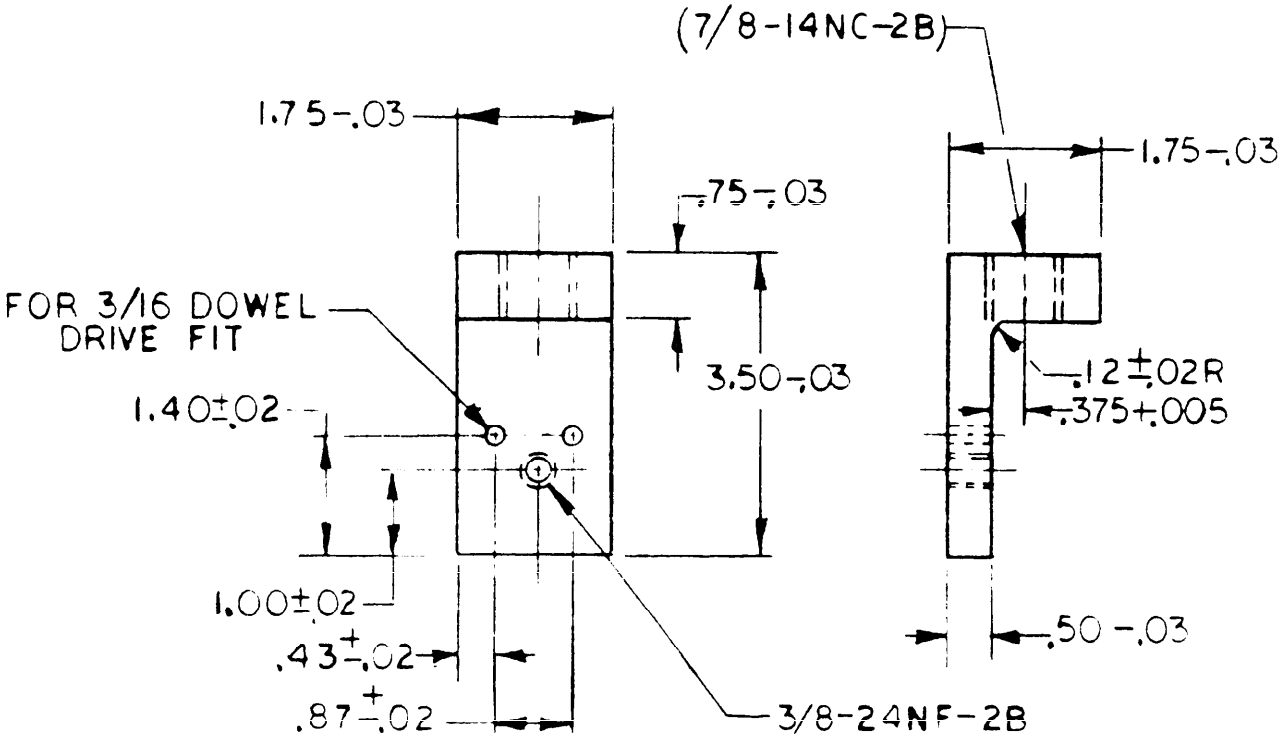


FIGURE 1.A OBTURATOR SPLIT DISC ASSEMBLY

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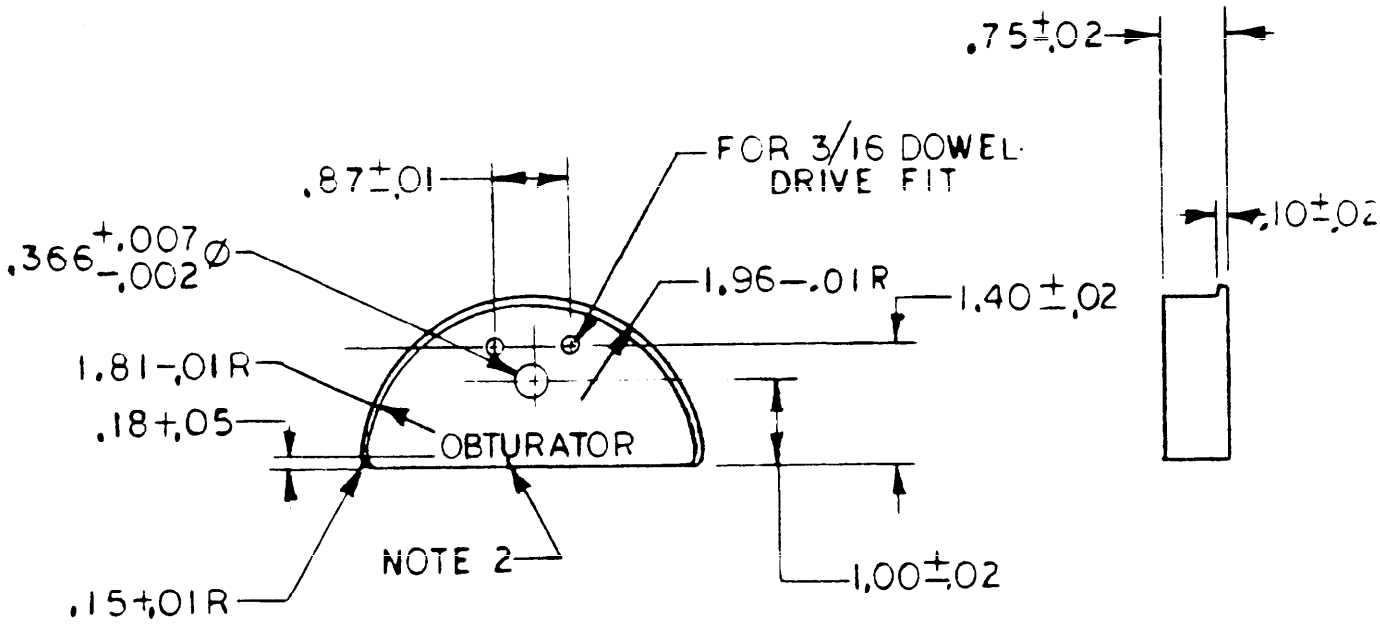


NOTES:-

1- MATERIAL:- STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D

FIGURE 1.B BASE SPLIT DISC (OBTURATOR)

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NOTES:—
1-MATERIAL:—STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D.
2-STAMP AS SHOWN WITH .25-.06 HIGH LETTERS.

FIGURE 1.C SPLIT DISC (OBTURATOR)

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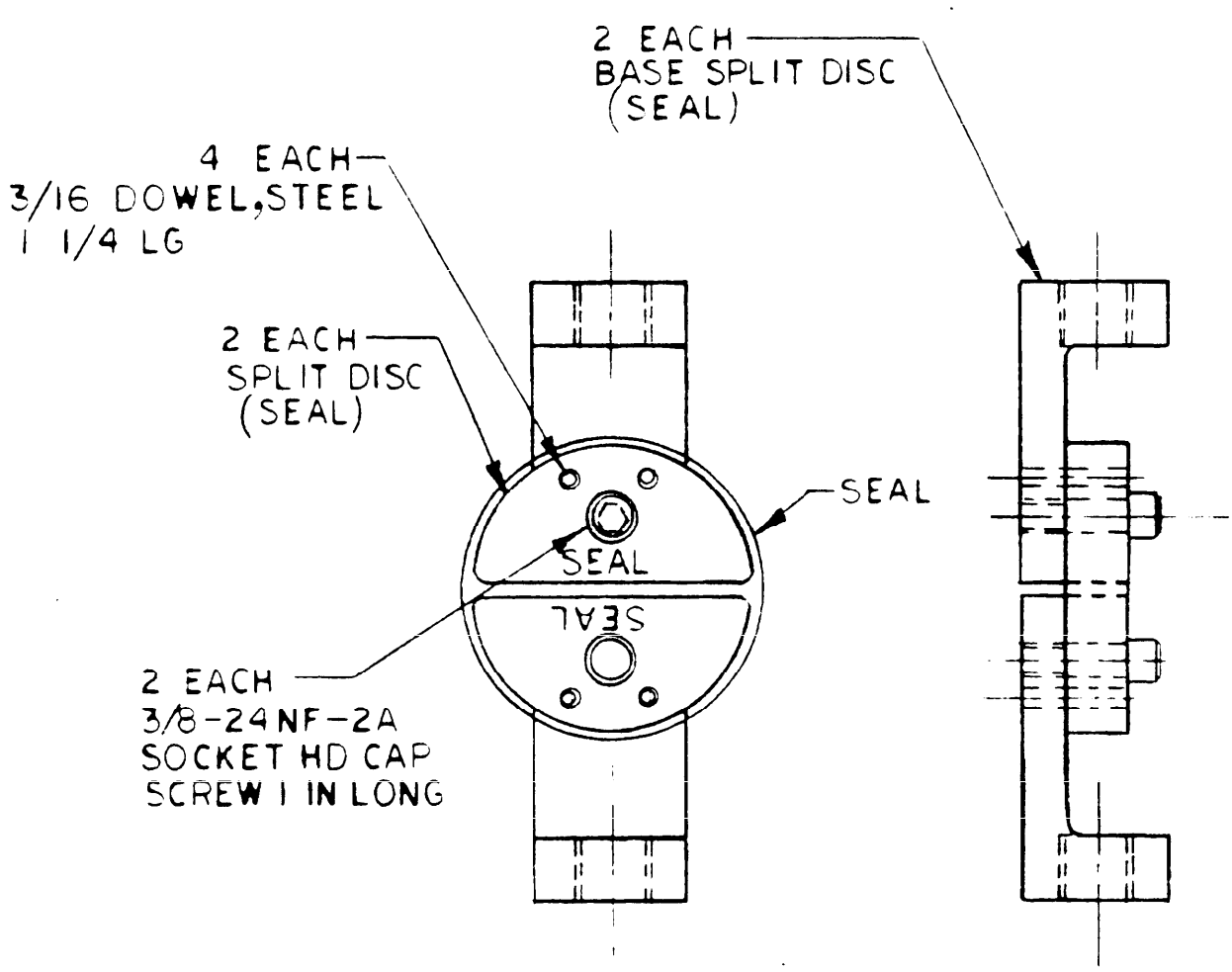
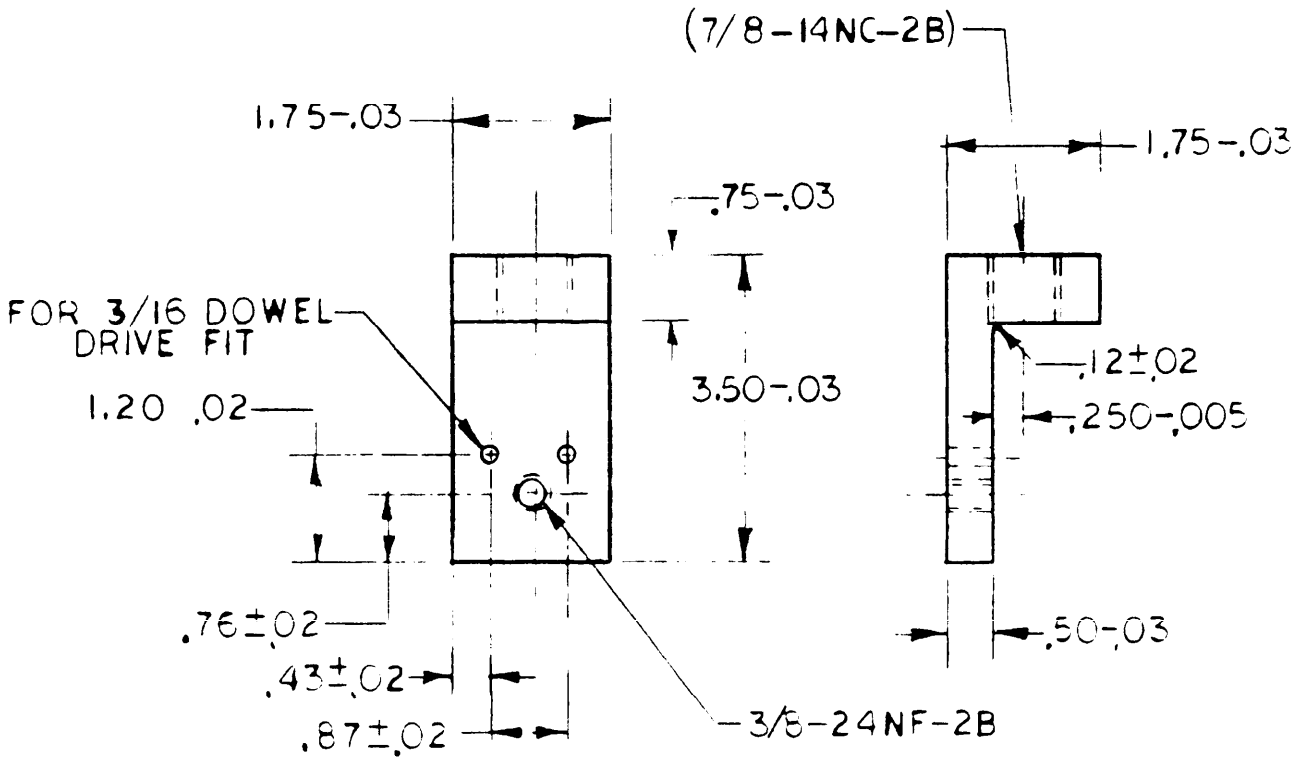


FIGURE 1.D SEAL SPLIT DISC ASSEMBLY

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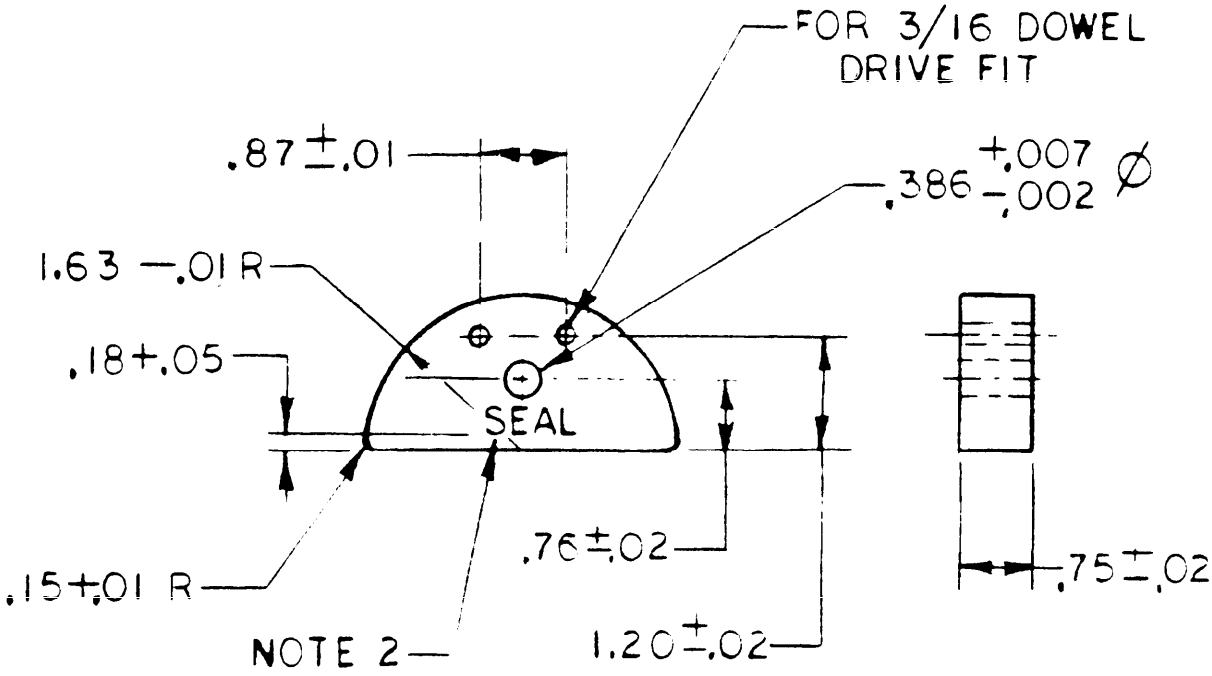


NOTES:-

1 MATERIAL STEEL, HOT ROLLED ALLOY BARS. GRADE 4340, ASTM-A322, 2 REQ'D.

FIGURE 1. E BASE SPLIT DISC (SEAL)

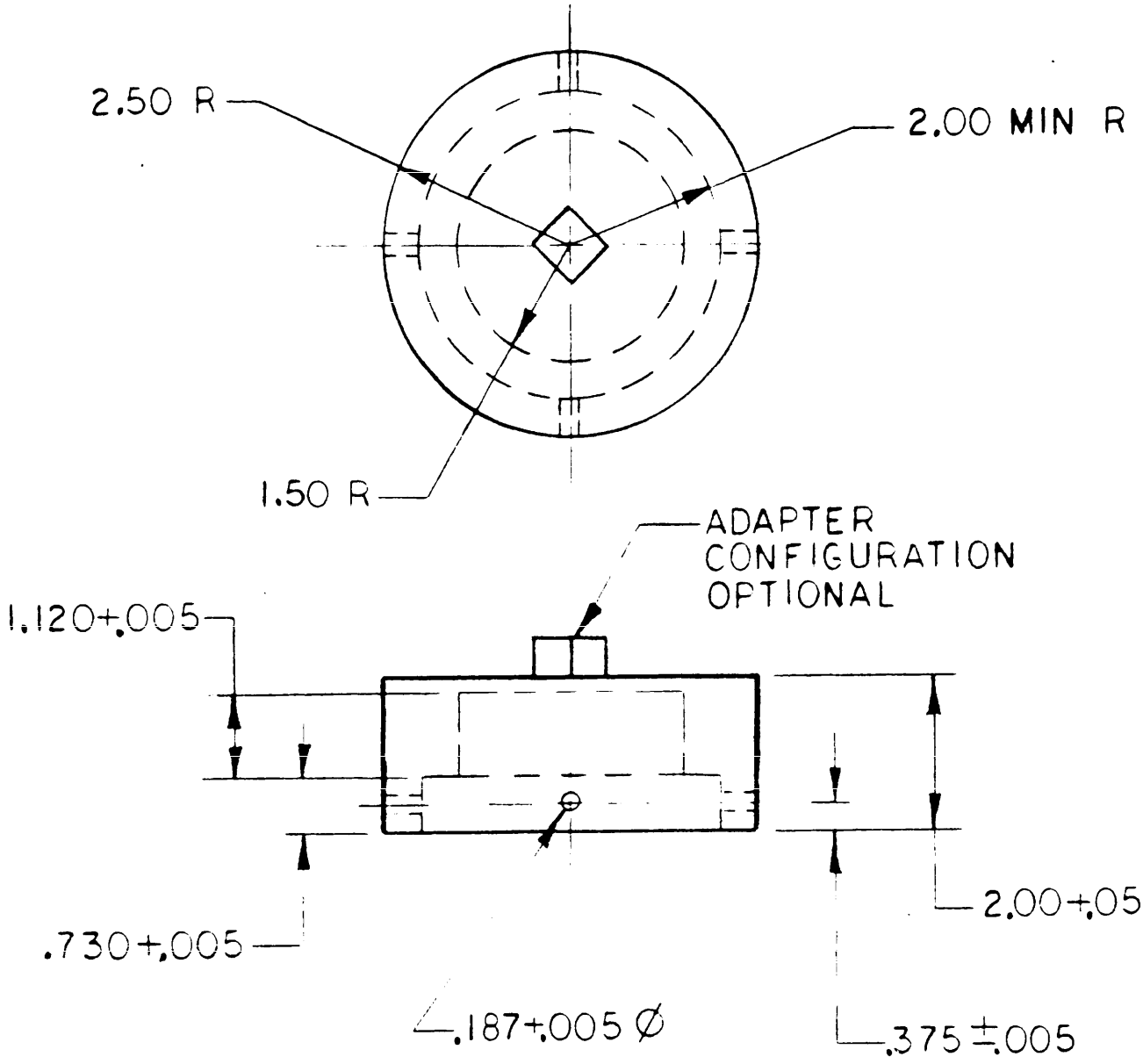
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NOTES:-
1-MATERIAL STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322, 2 REQ'D.
2-STAMP AS SHOWN WITH .25-.06 HIGH LETTERS.

FIGURE 1.F SPLIT DISC (SEAL)

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TORQUE FIXTURE
FIGURE 2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-60811B	PROJECTILE, 105MM, TP-T, M489 METAL PARTS ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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